Quantification: 876_z76c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	8
interior lacuna count	7
frame-edge lacuna count	1

Interior lacunae only: mean, median and sample SD (n = 7)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1619.29	1847	625.922	px^2
major_axis_length_px	lacuna shape	74.8552	85.0968	19.0124	px
minor_axis_length_px	lacuna shape	28.1986	28.6646	5.4419	px
aspect_ratio	lacuna shape	2.6446	2.6707	0.4723	unitless
eccentricity	lacuna shape	0.918	0.9273	0.0333	unitless
solidity	lacuna shape	0.8303	0.8641	0.0791	unitless
roots_count	headline	8.7143	9	3.7289	count
ring_length_r30_px	headline	568.143	602	136.215	px
ring_length_r60_px	reported	1614.14	1742	306.758	px
owned_length_px	ownership-dependent	6240.45	6421.94	1922.18	px
edge_count	ownership-dependent	458.714	424	150.377	count
mean_edge_length_px	ownership-dependent	13.6742	13.7296	0.8007	px
perimeter_px	normalised	231.415	249.12	45.1534	px
ring_area_r30_px2	normalised	8417.14	8971	1346.25	px^2
ring_area_r60_px2	normalised	22342.4	23523	2755.83	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0682888	0.0746909	0.0150752	px^-1
ring_density_r60	normalised	0.0729394	0.0768824	0.0145278	px^-1
roots_per_100px_perimeter	normalised	3.7413	3.7823	1.2071	per 100 px
ring_attached_length_r30_px	network v2	523.714	564	153.657	px
ring_attached_length_r60_px	network v2	1404.43	1402	432.808	px
ring_length_w_r30_px	network v2	639.047	672.274	151.542	px
ring_length_w_r60_px	network v2	1812.31	1933.41	342.603	px
sholl_crossings_r10	network v2	13.7143	14	3.4983	count
sholl_crossings_r20	network v2	17.2857	19	3.7289	count
sholl_crossings_r30	network v2	20.4286	21	5.2554	count
loop_count	loops, report only	62.4286	57	26.507	count
loop_edge_share	loops, report only	0.5334	0.5354	0.0763	share
ring_width_mean_r30_px	width	6.3371	6.4157	0.6145	px

Per field
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measure	value	units
analysed_area_px2	1.03371e+06	px^2
total_skeleton_length_px	66385	px
canalicular_length_density_per_px	0.06422	px^-1
junction_count	2237	count
junction_density_per_px2	0.00216405	px^-2
lacuna_count	8	count

lacunae_per_px2	7.74e-06	px^-2
skeleton_component_count	661	count
mean_component_length_px	100.431	px
median_component_length_px	13	px
field_length_density_w_per_px	0.0719805	px^-1
field_density_without_flagged_per_px	0.0657456	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	156	px
unattached_component_count	9	count
canalicular_length_density_attached_per_px	0.06422	px^-1
field_length_density_with_unattached_per_px	0.0643709	px^-1
loop_count	548	count
loop_edge_share	0.4218	share
edge_count	4995	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.1365	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	10	px
width_pixels_used	54367	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	3529	196.715	34.7861	5.655	0.9842	0.5697	-1.4629	16.8161	858.163	True
2	1841	89.0492	27.9031	3.1914	0.9496	0.8121	1.5494	146.123	381.449	False
3	2155	85.0968	33.783	2.5189	0.9178	0.8641	-1.5153	296.21	913.135	False
4	2003	88.5498	30.0304	2.9487	0.9407	0.8649	1.5644	297.345	325.757	False
5	1105	63.6459	23.8313	2.6707	0.9273	0.806	-1.5304	527.835	198.727	False
6	1847	70.5663	34.2892	2.058	0.874	0.8966	-1.5649	538.637	444.666	False
7	422	38.2927	18.8887	2.0273	0.8699	0.6709	-1.0621	646.408	104.687	False
8	1962	88.7858	28.6646	3.0974	0.9464	0.8975	-1.4695	687.621	605.365	False

Network per lacuna

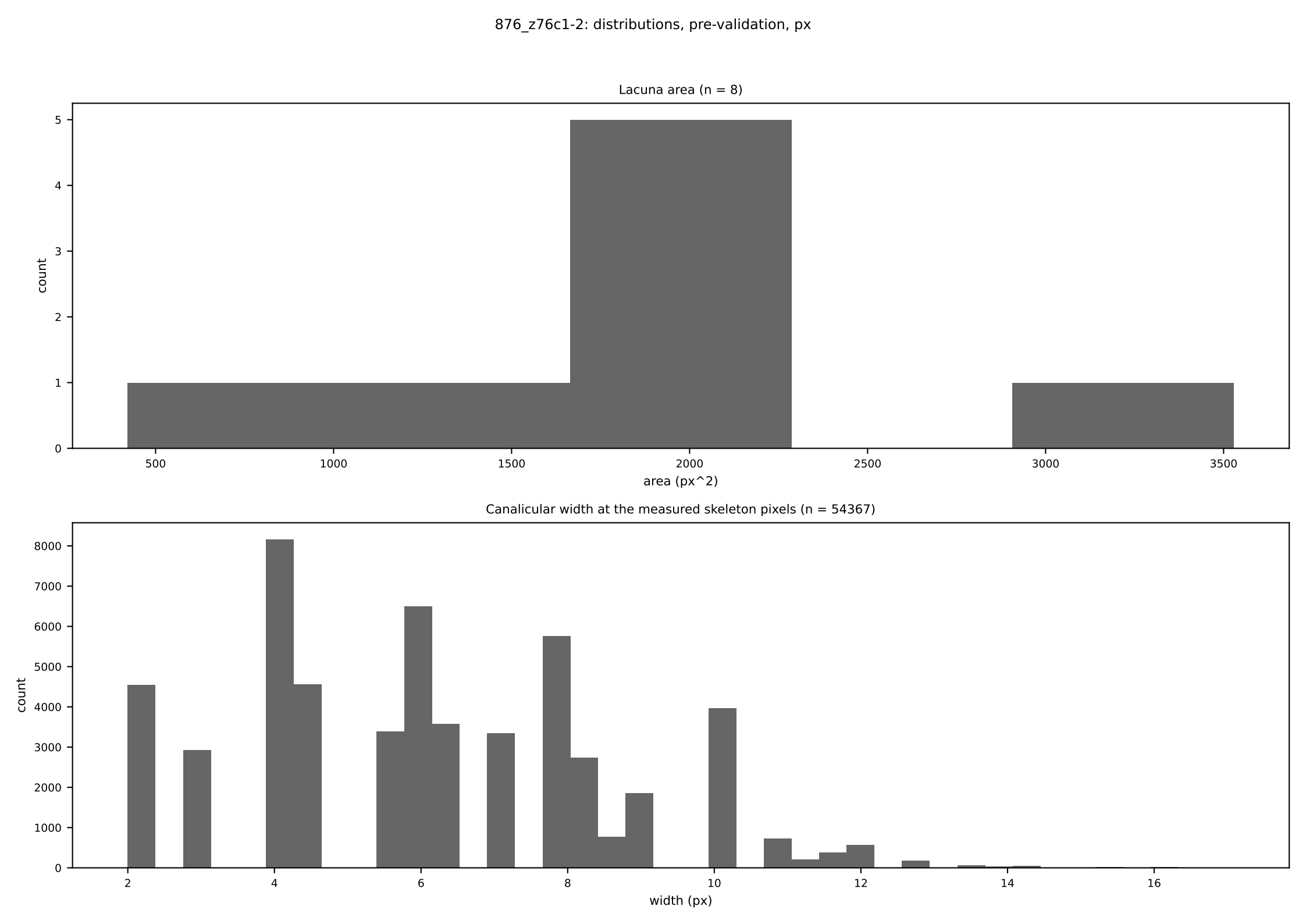
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	16	648	1549	5435.72	355	15.3119	3.688
2	16	666	1742	6421.94	424	15.1461	
3	9	447	1298	6789.25	485	13.9985	
4	10	766	1958	9578.03	724	13.2293	6.9134
5	6	602	1885	7131.32	564	12.6442	6.4157
6	9	645	1833	5642.86	411	13.7296	6.9445
7	6	435	1309	4190.91	319	13.1376	5.7621
8	5	416	1274	3928.82	284	13.8339	5.6496

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	756.927	11162	23083	0.6571	0.6039	0.0580541	0.0671057	2.1138
2	269.948	9365	24218	1	1	0.0711159	0.07193	5.9271
3	274.883	9487	24500	1	1	0.0471171	0.0529796	3.2741
4	264.392	9452	24487	1	1	0.0810411	0.0799608	3.7823
5	195.072	7582	20596	1	1	0.0793986	0.0915226	3.0758
6	210.593	8239	22047	1	1	0.0782862	0.0831406	4.2736
7	155.894	5824	17026	1	1	0.0746909	0.0768824	3.8488
8	249.12	8971	23523	1	1	0.0463716	0.0541598	2.0071

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	582	952	722.63	1720.24	17	16	19
2	637	1402	741.529	1933.41	18	21	21
3	402	915	495.99	1435.16	14	12	13
4	733	1838	867.37	2187.52	18	20	29
5	564	1862	672.274	2130.69	14	19	23
6	621	1797	722.073	2067.21	13	18	23
7	385	1107	502.617	1481.64	9	12	17
8	324	910	471.475	1450.53	10	19	17


876_z76c1-2: distributions, pre-validation, px
Lacuna area (n = 8)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 54367)
count
width (px)